8 The Elbow

Ariel A. Williams, Lance M. Brunton, Mark W. Anderson, and A. Bobby Chhabra

Abstract

The elbow is affected by a wide variety of degenerative, traumatic, inflammatory, and other Orthopaedic conditions, and magnetic resonance imaging (MRI) is often employed to guide diagnosis and treatment. This chapter presents a systematic approach to viewing and interpreting these studies. The role of MRI in evaluating common traumatic, degenerative and other elbow conditions is reviewed, and important MRI findings are described, with clinical correlation provided throughout.

Keywords: arthritis, magnetic resonance imaging, elbow abnormalities, elbow anatomy, elbow injuries

8.1 Normal Anatomy

8.1.1 Coronal Anatomy

The elbow comprises three joints: the radiocapitellar, ulnohumeral, and proximal radioulnar joints, all of which can be clearly visualized on the coronal view. Flexion and extension occur through the radiocapitellar and ulnohumeral joint whereas pronation and supination occur through the radiocapitellar and radioulnar joints.

The important ligamentous structures of the elbow are the medial and lateral collateral ligament (LCL) complexes (▶ Fig. 8.1). The medial collateral ligament (MCL) complex is seen on the coronal images as a vertically oriented, uniformly low-signal-intensity structure coursing between the medial epicondyle and the sublime tubercle. It has anterior, posterior, and transverse bundles. The anterior bundle is the most important stabilizer against valgus stress and is divided into the anterior band, the primary restraint in extension, and the posterior band, the primary restraint in flexion. The posterior bundle serves as a secondary restraint in flexion, while the transverse bundle makes minimal contribution to elbow stability. The LCL complex is a Y-shaped structure comprising the lateral ulnar collateral ligament (LUCL), the radial collateral ligament (RCL), and the annular ligament. There is some variability in normal anatomy of the LCL complex, and a fourth component, the accessory LCL, is also sometimes present.[1] The LUCLs and RCLs are seen on coronal images, both originating from the lateral epicondyle.[2] The LUCL descends obliquely from the lateral epicondyle to insert more posteriorly onto the ulna's supinator crest. The RCL comes down vertically to insert onto the annular ligament and supinator fascia. The annular ligament is best seen on axial and sagittal sections. By virtue of its Y-shape, the entire RCL complex contributes to lateral elbow stability.[3] However, the LUCL has long been felt to be the primary restraint to posterolateral rotatory instability.

The common extensor origin arises from the lateral epicondyle proximal to the LCL complex. It gives rise to the extensor carpi radialis brevis, extensor digitorum, communis, and extensor carpi ulnaris muscles. The extensor carpi radialis longus originates proximal to the common extensor origin at the distal supracondylar ridge. The common flexor origin arises from the medial epicondyle. It gives rise to the flexor carpi radialis, palmaris longus, flexor carpi ulnaris, flexor digitorum superficial humeroulnar head, and pronator teres. Distally, its fibers may appear confluent with the proximal MCL complex.

8.1.2 Sagittal Anatomy

The radiocapitellar and ulnohumeral joints are both observed on the sagittal images. The annular ligament can be appreciated as a ringlike structure encircling the radial head and stabilizing the proximal radioulnar joint.[4] On the posterior capitellum, a small depression is sometimes found. This "pseudodefect of the capitellum" is a normal variant and should not be confused for

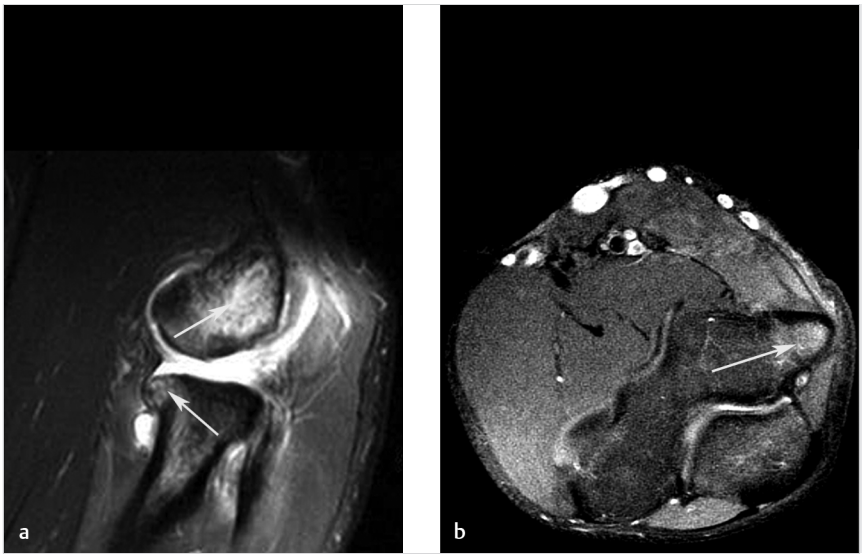


Fig. 8.1 Osseous injury. **(a)** Bone contusions. A sagittal short tau inversion recovery (STIR) image reveals high signal intensity within the capitellum and radial head, indicative of marrow edema (*arrows*). No discrete low signal fracture lines are evident. **(b)** Epiphysiolysis (Little Leaguer elbow). An axial STIR image of the right elbow showing diffuse high signal intensity within the medial epicondyle consistent with epiphysiolysis from repetitive valgus stress to the elbow (*arrow*). (Part **a**: Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.) (Part **b**: Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The Elbow. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:118–128.)

a chondral injury.[5] Although the ulnohumeral joint can also be seen on the coronal views, the articulation between the olecranon and olecranon fossa can be more clearly seen here on the sagittal as well as on the axial images.

The triceps tendon inserts onto the olecranon posterior to the elbow joint. It has three heads (medial, long, and lateral). On magnetic resonance imaging (MRI), the long and lateral head have a shared tendinous insertion that can be distinguished from the deeper and primarily muscular insertion of the medial head.[6] The biceps tendon inserts onto the radial tuberosity and onto the bicipital aponeurosis, a broad sheath of tissue that extends from the myotendinous junction to the antecubital fossa. Although the biceps tendon can usually be clearly identified on sagittal and axial images, the insertion into the bicipital aponeurosis can be difficult to visualize. The brachialis muscle can be seen deep to the biceps inserting onto the coronoid process and ulnar tuberosity.

8.1.3 Axial Anatomy

The axial sections allow excellent visualization of the radial head, the radioulnar joint, and the articulation between the olecranon and olecranon fossa. The common extensor and flexor origins are observed in cross-section as they arise from the lateral and medial epicondyles, respectively. The anconeus muscle is seen arising from the posterior aspect of the lateral epicondyle, coursing adjacent to the extensor carpi ulnaris and inserting onto the posterior ulna.[2] The triceps inserts onto the olecranon just proximal to the anconeus. Anteriorly, the biceps tendon inserts onto the radial tuberosity and the brachialis muscle lies deep to this. The brachialis's dual insertion is most clearly visualized on this view.[7]

The major nerves are best seen on the axial sections and can be traced along their lengths. The ulnar nerve lies posterior to the medial epicondyle where it courses through the cubital tunnel, the floor of which is formed by the MCL. It is the most easily visualized nerve due to its surrounding perineural fat. It then enters the forearm between the two heads of the flexor carpi ulnaris. The median nerve runs through the cubital fossa, medial to the brachial artery and biceps tendon. It then enters the forearm between the two heads of the pronator teres. The radial nerve travels adjacent to the lateral head of the triceps. It passes between the brachialis and brachioradialis, anterior to the lateral epicondyle, and after crossing the elbow joint, it gives rise to its terminal branches, the superficial sensory branch, and the posterior interosseous nerve.

8.2 Specialized Pulse Sequences and Imaging Protocols

MRI of the elbow has improved with the development of high-field systems and enhanced surface coil technology.[8] Most centers image the elbow with the patient in the supine position and the arm comfortably positioned at the patient's side, with the elbow in full extension. This position limits rotation of the proximal radioulnar joint with respect to the distal humerus and minimizes motion artifact. Alternatively, the patient can be positioned in the "Superman" position, prone with the arm over the head and the elbow extended. This position is less comfortable for the patient but places the elbow more centrally in the bore, which improves the signal-to-noise ratio and may sometimes be more feasible than supine positioning in larger patients who have difficulty fitting in the scanner.[9,10] Axial, coronal, and sagittal images are obtained routinely.

Most centers use a standard screening examination, consisting of an array of specified imaging planes and pulse sequences.[11] In addition to standard T1-weighted and T2-weighted images, fat-suppressed T2-weighted and short tau inversion recovery (STIR) sequences are extremely sensitive for detecting marrow and soft-tissue pathology by accentuating the increased signal of fluid and edema relative to the surrounding structures. These sequences are acquired with a fast spin-echo (FSE) technique that produces the desired contrast in a fraction of the time needed for conventional T2-weighted sequences. Gradient-echo sequences are often used to evaluate ligaments and articular cartilage (see Chapter 1, Essentials of MRI Physics and Pulse Sequences) and to delineate loose bodies in the elbow joint. However, these sequences should be avoided after elbow surgery (especially in the presence of metal hardware) because of the increased artifact inherent to the gradient-echo technique (see Chapter 1, Essentials of MRI Physics and Pulse Sequences). A proton-density pulse sequence is occasionally used, often with fat suppression, and combines characteristics of T1-weighted and T2-weighted images. FSE pulse sequences have substantially reduced the length of examination, a great benefit for claustrophobic patients. Imaging parameters can be refined additionally for optimal cartilage visualization when indicated (see Chapter 5, Articular Cartilage).[12]

MR arthrography is performed by injecting approximately 5 to 10 mL of diluted gadolinium contrast agent into the radiocapitellar joint from a lateral approach and acquiring fat-suppressed T1-weighted and T2-weighted images.[13] Although somewhat controversial, MR arthrography of the elbow may help evaluate undersurface collateral ligament tears, capsular disruption, osteochondritis dissecans (OCD), and intra-articular loose bodies. When investigating infectious, neoplastic, or synovial disorders involving the elbow, intravenous gadolinium is often used to evaluate for synovial hypertrophy or enhancement of pathologic tissues. Intravenous contrast is also useful for detecting soft tissue abscesses, which show a thick, enhancing wall surrounding the nonenhancing central fluid. Additional refinement of conventional MRI pulse sequences may eventually render these invasive techniques obsolete.[14]

8.3 Systematic Approach to Image Review

Coronal, sagittal, and axial views should all be examined including T1, T2, and any specialized sequences. T2-weighted images are reviewed first to look for abnormal regions of increased T2-weighted signal intensity. Findings from T2-weighted images are then correlated with T1-weighted images to allow better evaluation of anatomic detail. A systematic, structure-by-structure approach is utilized to evaluate for specific pathology. The bony structures are evaluated for the presence of stress reactions, occult fractures, evidence of infection or neoplasia, or, in the skeletally immature, apophyseal or physeal injury. Joint congruity is evaluated and osteophytes and joint space collapse are noted. The cartilage and subchon-

dral bone are evaluated for localized or diffuse chondral injuries, OCD lesions, or erosive change. The synovium and synovial fluid should also be examined. The presence of an effusion should be noted, as well as any loose bodies or suggestion of atypical synovial proliferation. The ligamentous structures are inspected carefully. Abnormal signal intensity or frank discontinuity is strongly suggestive of partial or complete ligament injury. MR arthrogram may also show contrast extravasation. The common flexor and extensor origins are evaluated for tendinosis or tearing. Similarly, the biceps, triceps, and brachialis tendons are examined to look for abnormal signal intensity and partial or complete rupture. The adjacent soft tissues should be examined for abnormal fluid collections or masses. Finally, the major nerves should be traced down their length to evaluate for compression or signal changes.

8.4 Traumatic Conditions

Most patients with elbow trauma may be evaluated and treated adequately without the use of advanced imaging. However, for select patients, MRI may facilitate the diagnosis of suspected occult fractures, loose bodies, ligamentous injuries, and tendinous pathology.

8.4.1 Occult Fractures

Conventional radiographs often are unrevealing in a patient presenting with elbow pain after acute trauma. In this setting, a joint effusion is often detected clinically, and an elevated posterior fat pad sign may be evident on conventional radiographs. Prospective studies in adults and children have shown that the presence of a posterior fat pad sign correlates with an occult fracture in more than 75% of patients.[15,16] However, localizing the injury is a challenging endeavor, and patients often may be overtreated with immobilization in the absence of advanced imaging.

MRI may be used in the setting of elbow trauma to localize and characterize osseous injury when conventional radiographs are unrevealing or equivocal. Fat-suppressed T2-weighted (or STIR) pulse sequences are the most sensitive images for detecting radiographically occult traumatic or stress fractures. Fractures show a linear pattern of signal change on T1-weighted (decreased signal) or T2-weighted (increased signal) images, whereas an osseous contusion often has a more nonspecific appearance (▶ Fig. 8.1a). The most frequent site of occult fracture is the radial head; other sites, such as the lateral epicondyle or olecranon, are much less common.[16] The radial head fractures may be accompanied by associated injuries that are most effectively detected by MRI, such as ligament or capsular disruptions, OCD, loose bodies, or capitellar bone contusions.[17] Olecranon stress fractures can be seen in throwing athletes, possibly as a result of valgus extension overload, and may be accompanied with associated injuries such as MCL injuries and, in skeletally immature patients, medial epicondyle avulsion fractures.[18]

In children with a traumatic injury to the elbow and radiographic evidence of only a joint effusion, MRI is a sensitive and accurate modality for identifying or excluding an occult fracture.[19] MRI reveals a broad spectrum of elbow pathology in this setting, including the following[20]:

- Bone bruises.
- Ligamentous disruptions.
- Muscle injury.

MRI also may be effective in evaluating pediatric physeal injuries, especially in young children with poorly visualized ossification centers.[21] Finally, MRI has a role in the detection of Little Leaguer elbow, a valgus stress injury in skeletally immature throwing athletes in which epiphysiolysis occurs at the medial epicondylar apophysis (▶ Fig. 8.1b).[22]

8.4.2 Loose Bodies

The clinical manifestation of loose bodies within the elbow joint is typically pain with motion, often accompanied by mechanical symptoms, such as catching or locking. Loose bodies may originate from a traumatic shear injury to the osteochondral surface, synovial disorders, or OCD. The definitive treatment of symptomatic loose bodies is arthroscopic or open removal, but accurate diagnosis is imperative to prevent unnecessary surgery.

Although conventional radiographs or CT may detect the presence of osseous loose bodies in the elbow joint, MRI is especially useful for detecting intra-articular cartilaginous or osteocartilaginous loose bodies. Loose bodies are found most commonly in the olecranon or coronoid fossae on T2-weighted images (▶ Fig. 8.2). Osteophytes and synovial hypertrophy may mimic loose bodies on MRI images.[11] The use of MRI for detecting loose bodies in the elbow has been disputed, and an overall

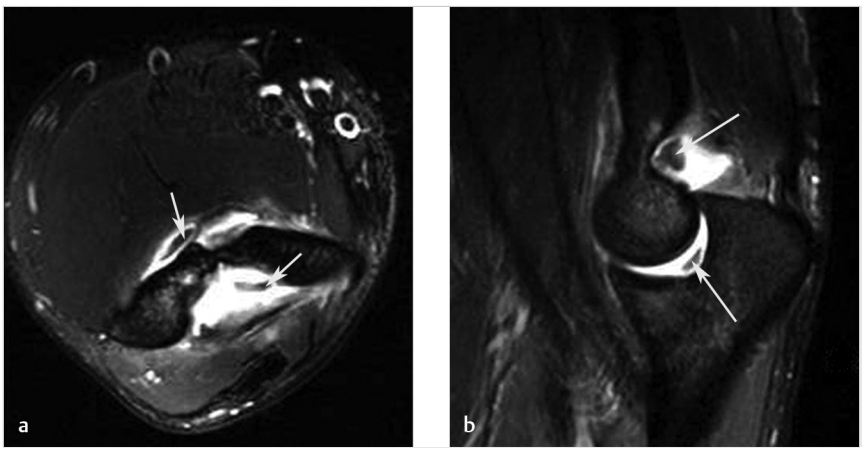


Fig. 8.2 Axial **(a)** and sagittal **(b)** T2-weighted images showing loose bodies in the coronoid and olecranon fossae and in the trochlear ulnar joint (*arrows on each*). (Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.)

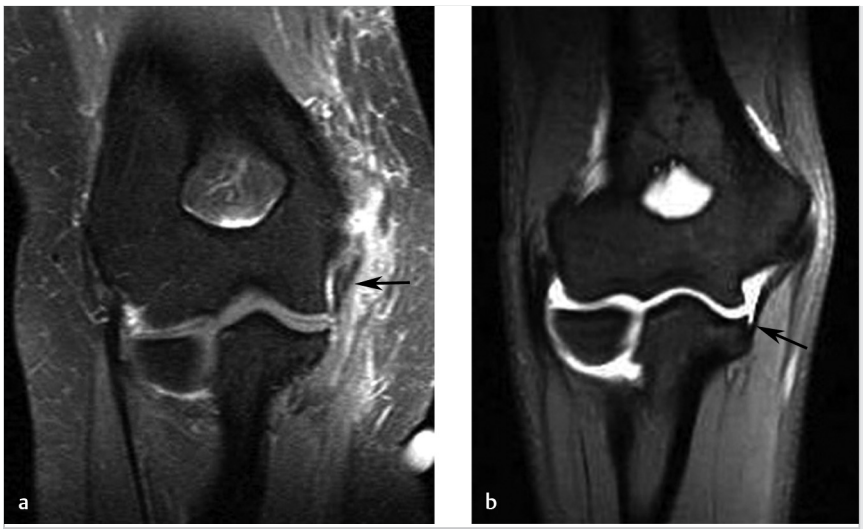


Fig. 8.3 Medial collateral ligament (MCL) injury of the right elbow. **(a)** A coronal fat-suppressed T2-weighted image showing discontinuity (complete tear) of the MCL with surrounding edema and hemorrhage (*arrow*). **(b)** A coronal fat-suppressed T1-weighted arthrogram showing a partial tear of the MCL as a "T-sign" with leakage of intra-articular contrast at the undersurface of the MCL at its distal insertion on the sublime tubercle (*arrow*). (Part **a**: Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.) (Part **b**: Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The Elbow. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:118–128.)

low specificity and sensitivity, similar to that of conventional radiography, when compared with arthroscopic findings have been reported.[23] MR arthrography may be effective in this setting, but it has not been studied adequately.

8.4.3 MCL Injury

MCL injuries have received tremendous attention because of their prevalence in throwing athletes of all ages and at all levels of competition. Accurate diagnosis of a full-thickness or partial-thickness tear of the MCL has important implications for treatment, as does the differentiation between MCL pathology and other causes of medial-side elbow pain, such as ulnar neuropathy, stress fracture, and flexor-pronator mass injury.

Full-thickness tears of the anterior bundle of the MCL can be detected accurately by MRI. The abnormalities detectable by MRI include ligament attenuation, redundancy, or discontinuity. Increased signal intensity often is found within and adjacent to the ligament on fat-suppressed T2-weighted images, indicating edema or hemorrhage (▶ Fig. 8.3a).[14] MRI is less reliable for detecting partial-thickness tears, but the addition of intra-articular contrast may improve its sensitivity (▶ Fig. 8.3b).[24,25] Other findings related to chronic MCL insufficiency are not unusual. These include adjacent muscle edema, ulnar neuritis, and chondral injuries of the radiocapitellar joint. Signs of posteromedial elbow impingement may also be present, including posterior trochlea and anterior olecranon chondrosis and marginal osteophytes of the trochlea and olecranon.[14,26]

8.4.4 LCL Complex Injury

Injury to the lateral ligamentous complex from an acute elbow dislocation may lead to chronic mechanical symptoms or recurrent instability when early treatment is inadequate. Tears of the LUCL and RCL are common in acute elbow dislocations. Chronic insufficiency of the LUCL is believed to lead to posterolateral rotatory instability, which is often difficult to diagnose clinically.[27]

The ability of MRI to visualize tears of the LCL complex has been historically less dependable, especially in patients with chronic instability.[28] One study of asymptomatic individuals revealed inconsistent MRI signal characteristics within the

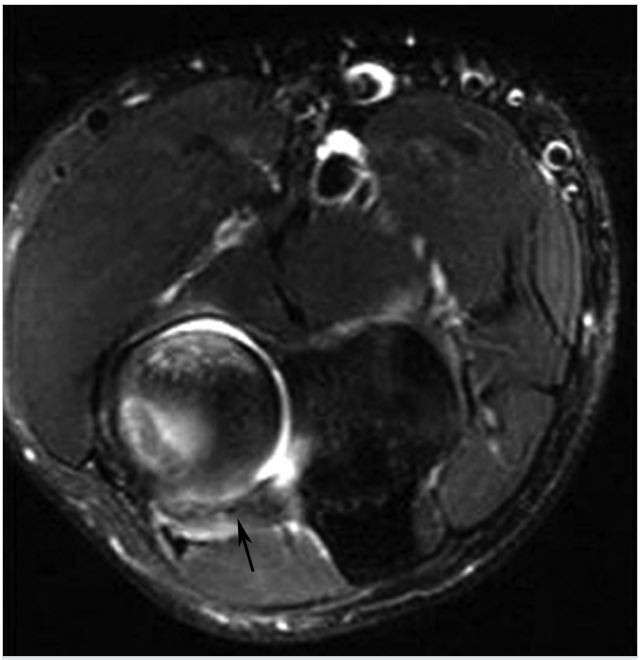


Fig. 8.4 An axial T2-weighted image of the right elbow showing thickening and abnormal signal (sprain vs. partial tear) of the lateral ulnar collateral ligament (LUCL) with adjacent high signal intensity edema posterior to the radial head (*arrow*). (Adapted with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.)

LUCL.[29] Other investigators, however, have found MRI to be effective in detecting abnormalities of the LUCL in the setting of posterolateral rotatory instability when 3D gradient-echo and FSE sequences were used.[30] Findings are similar to those seen in MCL injuries, with combinations of ligament attenuation, redundancy, and/or discontinuity evident on coronal or axial images (▶ Fig. 8.4).[21] MR arthrography may show contrast extravasation through complete LUCL tears but has not been well studied in this setting.[31] MRI also demonstrates increased radiocapitellar and ulnohumeral joint incongruity in some patients with instability.[32]

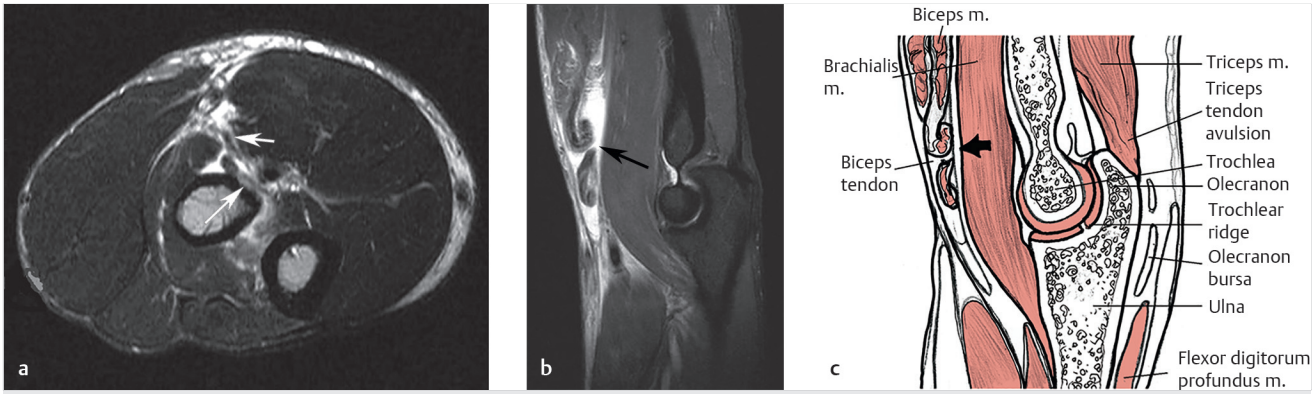


Fig. 8.5 Distal biceps tendon injury. **(a)** An axial T2-weighted image of the right elbow showing the ruptured and retracted distal biceps tendon (*arrows*) with associated surrounding edema and hematoma. **(b)** A sagittal T2-weighted image showing a complete tear of the biceps tendon with prominent edema and hemorrhage surrounding the thickened, retracted tendon (*arrow*). **(c)** Artist's sketch. (Part **a**, **c**: Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The Elbow. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:118–128.) (Part **b**: Reproduced with permission from Magnetic Resonance Imaging of the Elbow: Update on Current Techniques and Indications Brunton, Lance M. et al. Journal of Hand Surgery, Volume 31, Issue 6, 1001–1011)

8.4.5 Biceps Tendon Injury

Injury to the distal biceps tendon is relatively common. Complete rupture occurs almost exclusively in men after forced extension of the flexed elbow against an eccentric load. However, symptoms arising from a partial tear may have an insidious onset, and such tears usually occur from repetitive microtrauma in an already degenerative tendon. Although MRI is not always necessary to diagnose a complete distal biceps tendon rupture, it can distinguish between tendinosis, partial tears, and complete tears. MRI may also allow for the diagnosis of concomitant injuries. A complete rupture is characterized by the absence of the low signal intensity tendon tissue at the radial tuberosity insertion on axial images. The less experienced clinician may have difficulty localizing the biceps tendon insertion on the axial images and may find it easier to make or confirm the diagnosis on the sagittal T2-weighted images. A large amount of high signal intensity fluid and/or hemorrhage in the antecubital fossa usually accompanies an acute distal biceps tendon injury. The extent of tendon retraction is assessed on sagittal images, although the course of the tendon often is oriented obliquely to this plane (▶ Fig. 8.5). An abnormally thickened or attenuated distal biceps tendon with increased signal intensity may represent tendinopathy or a partial tear. Associated bone marrow edema within the radial tuberosity is likely related to microavulsive injury. Increased fluid within the bicipitoradial bursa may accompany a partial tear in some patients.[21,33]

8.4.6 Triceps Tendon Injury

Ruptures of the triceps tendon are far less common than are injuries to the distal biceps tendon. Complete tears result from a traumatic event and are manifested clinically by the inability to extend the elbow against gravity. Extensive posterior elbow swelling, ecchymosis, and a palpable tendon defect are also common findings. The tendon typically is detached from its insertion on the olecranon process of the proximal ulna, and a small osseous avulsion may be seen on conventional radiographs.

MRI has a role in diagnosing triceps tendon injury in patients with equivocal clinical findings. A complete rupture is accompanied by extensive adjacent soft tissue edema, and the retracted tendon margin is seen on sagittal images (▶ Fig. 8.6). Partial injury or tendinopathy is more common and may result only in thickening or thinning of the tendon, sometimes without related abnormal signal intensity.[34] Findings of olecranon bursitis may accompany triceps tendon injury in some patients.

8.5 Degenerative Conditions: Lateral and Medial Epicondylitis

The term *epicondylitis* may be a misnomer because mucoid degeneration and tearing of the extensor carpi radialis brevis tendon (in the case of lateral epicondylitis), rather than inflammation, are the key histologic features of the disorder.[35]

Nevertheless, in the absence of a more accurate term, lateral epicondylitis (also known as *tennis elbow*) is the most common overuse disorder of the elbow, occurring most frequently in individuals who participate in racquet sports and in middle-aged persons who perform repetitive wrist extension in their daily activities. Patients complain of lateral elbow pain that increases with the offending activity. Examination reveals tenderness over the insertion of the common extensor tendon at the lateral epicondyle, and the pain is exacerbated by resisted wrist extension with the elbow fully extended. Although the diagnosis usually is obvious by history and physical examination, concurrent underlying pathology cannot be ruled out easily. The differential diagnosis for lateral elbow pain includes the following:

- Degenerative joint disease.
- OCD.
- Radial tunnel syndrome.
- Posterolateral rotatory instability.
- Occult fracture.
- Loose bodies.

It is important to consider these other etiologies when nonoperative treatment of lateral epicondylitis for more than 6 to 12 months has failed.

Usually, MRI is used to rule out concomitant causes of refractory elbow pain rather than to diagnose lateral epicondylitis.[36] The typical MRI finding of isolated lateral epicondylitis is

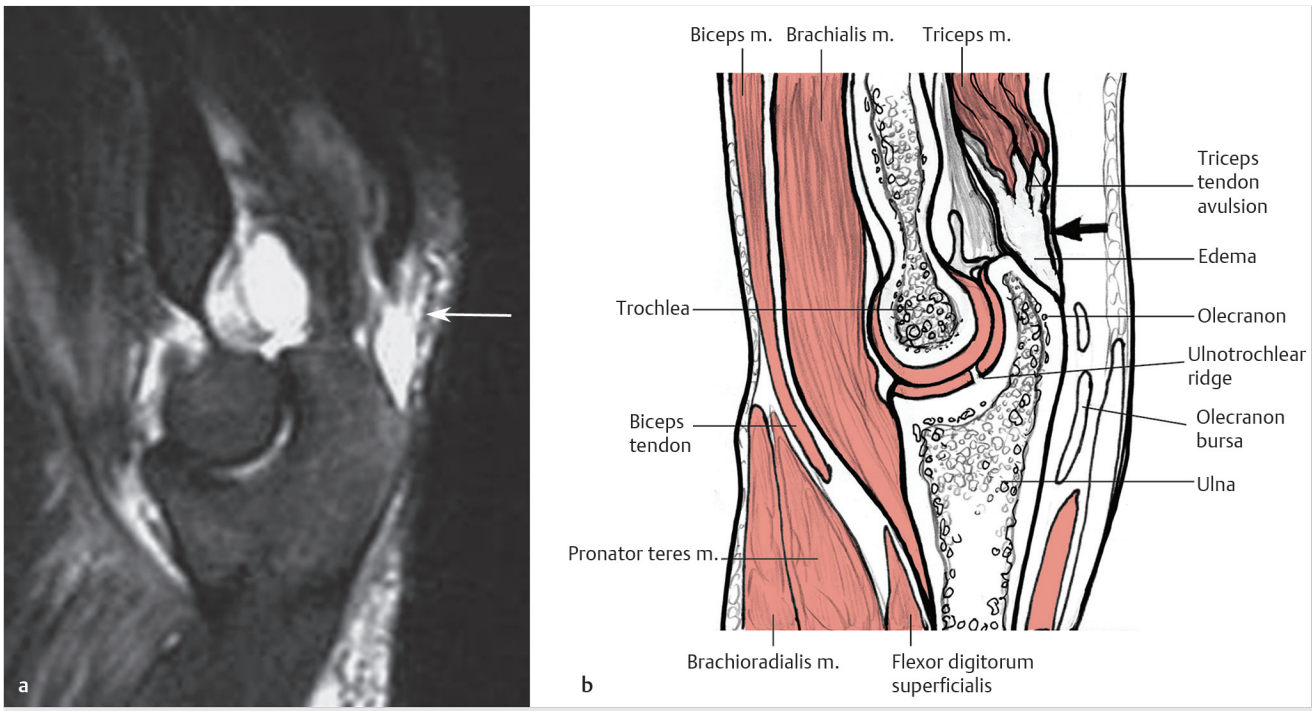


Fig. 8.6 A sagittal T2-weighted image **(a)** and artist's sketch **(b)** revealing a torn, retracted triceps tendon with a large amount of high-signal-intensity edema between the olecranon process and the tendon end (*arrows on each*). (Part **a**: Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.) (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The Elbow. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:118–128.)

increased signal intensity and thickening of the common extensor tendon origin, in some cases with adjacent soft tissue edema (▶ Fig. 8.7).[21,37] In patients with severe lateral epicondylitis, the common extensor tendon origin may be separated off the lateral epicondyle.[36] Abnormalities of the LCL complex, especially the LUCL, are also often seen (▶ Fig. 8.8).[38] Increased signal intensity and thickening of the common extensor tendon origin can be incidental findings in some patients without symptoms of lateral epicondylitis, especially in older individuals, but in these cases soft tissue edema and other related findings are usually absent.[39]

Similar to lateral epicondylitis, the diagnosis of medial epicondylitis can usually be made clinically, and the primary role of MRI is to rule out other diagnoses, such as MCL injury or ulnar neuritis, in patients for whom nonoperative treatment fails. The MRI findings in medial and lateral epicondylitis are similar. Medial epicondylitis is characterized by a thickened common flexor tendon and local soft tissue edema.



Fig. 8.7 A coronal T2-weighted image of the right elbow showing concurrent severe lateral epicondylitis (*upper arrow*) with high signal intensity within a thickened common extensor tendon and discontinuity of the underlying RCL (*lower arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The Elbow. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:118–128.)

8.6 Other Pathologic Conditions

8.6.1 Infections

Depending on the acuity of presentation, various imaging modalities are used in the evaluation of patients with signs and symptoms of musculoskeletal infection. Conventional radiographs, computed tomography (CT), and radioisotope studies all have roles in the diagnosis of an infectious process, although MRI is the single most useful study for determining the precise anatomic location and extent of infection.

Acute osteomyelitis causes inflammation of medullary bone, which can be detected by MRI well before findings are evident on conventional radiographs. On T1-weighted images, areas of

marrow involvement show a low signal intensity that contrasts with the surrounding bright normal marrow. STIR images, or other fat-suppressed T2-weighted images, are the most sensitive for infection, showing affected areas as high signal intensity. Sometimes these regions of high T2-weighted signal are accompanied by a well-defined low-intensity margin. Because these findings may be similar in patients with tumor or early osteonecrosis, correlation with the history and physical examination is critical.[40]

The role of MRI in patients with presumed septic arthritis is less clear. Although the presence of an elbow joint effusion can be well depicted by MRI, characterizing the synovial fluid as normal, bloody, or purulent is less reliable. MRI often is used to determine the extent of involvement in patients with severe cellulitis, septic bursitis, or other superficial infections that are unresponsive to nonoperative measures. MRI also is an excellent method for diagnosing a suspected deep abscess about the elbow that may require surgical incision and drainage.[41]

The MRI findings in septic and nonseptic olecranon bursitis may be markedly similar. Abnormalities may include the following:

- Marginal lobulation.
- Bursal septation.
- Heterogeneous internal signal.
- Poorly defined margins.
- Adjacent soft tissue edema.
- Thickening or edema of the triceps tendon.
- Localized bone marrow edema.

The most consistent finding in septic olecranon bursitis is rim and soft tissue enhancement on postgadolinium T1-weighted images.[42]

8.6.2 OCD

OCD of the humeral capitellum is an idiopathic disorder affecting adolescents with a history of repetitive overuse of the elbow. Affected individuals commonly present with pain and swelling of the lateral aspect of the elbow. Mechanical symptoms and/or limitations of motion may be indicative of an associated loose body. MRI can be helpful in the early detection of OCD, when conventional radiographs may show only subtle changes. Focal areas of subchondral low signal intensity on T1-weighted images are characteristic. Fat-suppressed T2-weighted images provide the most sensitive evaluation for a potential OCD lesion at the capitellar articular surface; however, these images are the least specific. With a patient in whom the clinical suspicion is relatively high, the fat-suppressed T2-weighted image can be used as a "screening" examination and then correlated with the T1-weighted image where anatomic detail may be better seen. MRI can also help to evaluate the stability and viability of the OCD fragment. Fluid surrounding the fragment on fat-suppressed T2-weighted images or surrounding contrast material on MR arthrography is a sign of an unstable fragment (▶ Fig. 8.9).[43] Enhancement of a fragment after intravenous gadolinium



Fig. 8.8 A coronal short tau inversion recovery (STIR) image of the right elbow showing isolated lateral epicondylitis, indicated by the thickened common extensor tendon with amorphous features and high signal intensity (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.)



Fig. 8.9 A coronal oblique short tau inversion recovery (STIR) image of the right elbow showing an unstable osteochondritis dissecans (OCD) fragment of the capitellum with joint fluid tracking behind the fragment (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.)

contrast has been administered suggests adequate blood supply and fragment viability.[44]

8.6.3 Compression Neuropathies

The most commonly occurring compression neuropathies about the elbow include the following:
- Cubital tunnel syndrome (ulnar nerve).
- Radial tunnel syndrome (posterior interosseous nerve—pain).
- Posterior interosseous nerve syndrome (posterior interosseous nerve—motor deficits).
- Pronator syndrome (median nerve).

Each syndrome has key historical and physical examination findings that have been addressed elsewhere.[45] Electromyography and nerve conduction studies are the most frequently used adjunctive diagnostic tests. Advanced imaging may have a role in evaluating patients for whom nonoperative treatment fails or those with recurrence after surgical intervention.

MRI is most useful for assessing cubital tunnel syndrome, especially when electrodiagnostic testing is equivocal, or for excluding accompanying pathology in patients in whom this syndrome recurs.[46,47] MRI may be more sensitive than nerve conduction studies in the diagnosis of ulnar neuropathy at the elbow.[48] Prolonged compression of peripheral nerves leads to a spectrum of MRI findings, including the following:
- Thickening of the nerve.
- Increased signal intensity within the nerve.
- A mottled appearance of the nerve.
- A surrounding fluid collection, which may represent varying degrees of inflammatory changes (▶ Fig. 8.10).

The median and radial nerves are more difficult to evaluate; however, MRI can detect adjacent space-occupying lesions that may produce compression. Patients with chronic neuropathy also may show changes within adjacent denervated muscles, such as fatty atrophy on T1-weighted images (▶ Fig. 8.11) or muscle edema on T2-weighted images.[49]

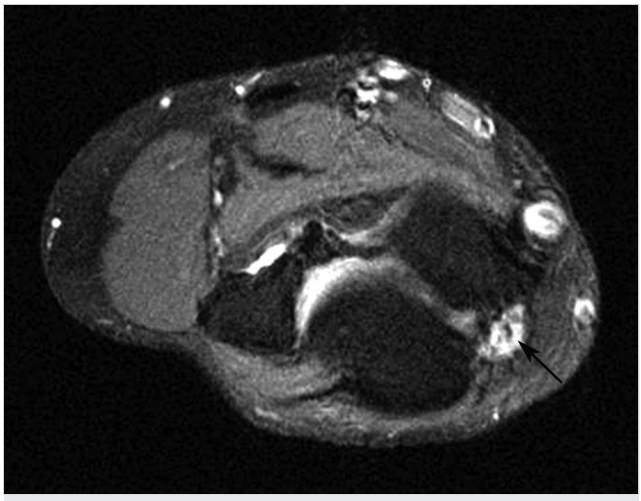


Fig. 8.10 An axial short tau inversion recovery (STIR) image of the right elbow showing a thickened ulnar nerve with a mottled appearance and surrounding increased signal intensity consistent with edema from inflammation (ulnar neuritis) (*arrow*). (Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.)

8.6.4 Synovial Disorders

MRI can depict numerous abnormal synovial-based processes, such as rheumatoid arthritis (RA), crystal deposition disorders, pigmented villonodular synovitis (PVNS), and idiopathic synovial osteochondromatosis.[50] When synovial fluid analysis is equivocal, MRI may aid in differentiating conditions that produce a joint effusion. With the use of intravenous gadolinium, MRI has become a powerful tool for detecting the early stages of inflammatory arthritides, such as RA, because of the striking enhancement of the proliferative synovium. Although PVNS and synovial chondromatosis are commonly seen in the hip and knee and are quite rare in the elbow, these syndromes should be considered in the differential diagnosis of the patient with multiple loose bodies in the elbow. Gradient-echo imaging shows accentuated "signal dropout" in areas of PVNS, relating to the presence of hemosiderin in this disease process. Other nonspecific MRI findings of PVNS include variable synovial proliferation, bone erosion, and joint effusion.[51,52]

8.6.5 Soft Tissue Masses

Although soft tissue masses are less common around the elbow than in the lower extremity, MRI is essential for evaluating such lesions. The primary role of MRI is differentiating determinate from indeterminate lesions. Determinate lesions are those for which a well-trained musculoskeletal radiologist or clinician can provide a single diagnosis after MRI evaluation with a high degree of certainty based on the imaging appearance. Such lesions include the following[21,53,54]:
- Lipomas.
- Ganglions.
- Hemangiomas.

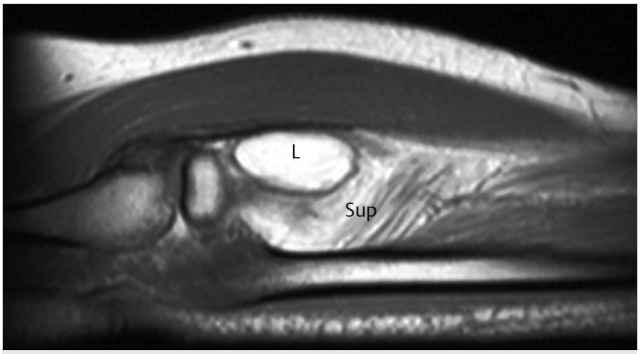


Fig. 8.11 A sagittal T1-weighted image revealing a lipoma (*L*) within the supinator muscle (*Sup*). The supinator shows diffuse high signal intensity, which matches that of the lipoma and subcutaneous fat, indicating fatty atrophy of the denervated muscle from long-standing compression of the posterior interosseous nerve by the lipoma. (Reproduced with permission from Brunton LM, Anderson MW, Pannunzio ME, Khanna AJ, Chhabra AB. Magnetic resonance imaging of the elbow: update on current techniques and indications. J Hand Surg Am 2006;31:1001–1011.)

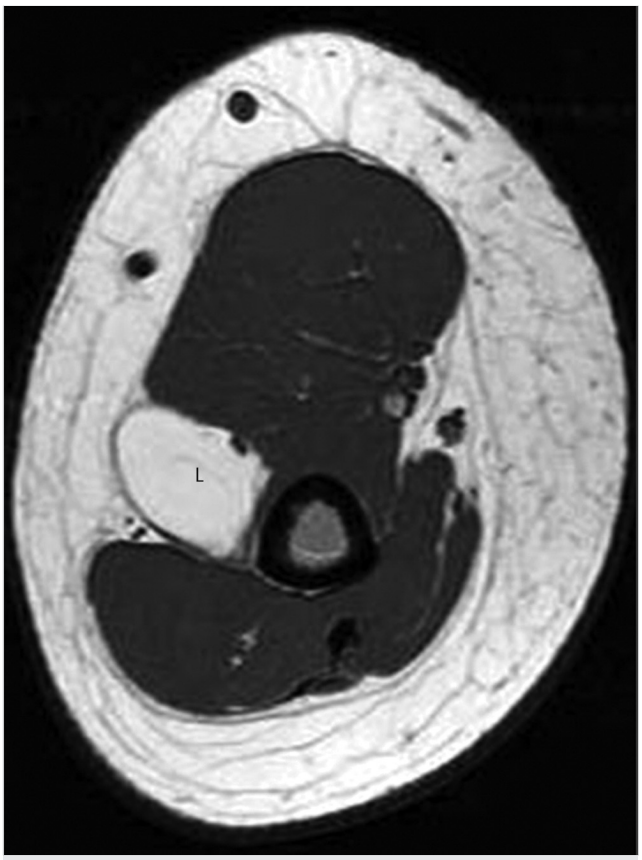


Fig. 8.12 An axial T1-weighted image showing the similarity in signal intensity between the well-circumscribed lipoma (*L*) and the adjacent subcutaneous fat. (Reproduced with permission from Brunton LM, Anderson MW, Chhabra AB. The Elbow. In: Khanna AJ, ed. MRI for orthopaedic surgeons. New York: Thieme; 2010:118–128.)

- Neurofibromas.
- Muscle tears.
- Myositis ossificans (chronic stage).
- PVNS.

Determinate lesions are those for which a well-trained musculoskeletal radiologist or clinician can provide a single diagnosis after MRI evaluation with a high degree of certainty based on the imaging appearance. Such lesions include the following: Lipomas. Ganglions, Hemangiomas. Neurofibromas. Muscle tears. Myositis ossificans (chronic stage). PVNS. Lipomas are amongst the most commonly encountered determinate lesions. A lipoma matches the signal intensity of subcutaneous fat on all pulse sequences, including fat-suppressed sequences (▶ Fig. 8.12). Indeterminate lesions are those lesions in which the radiologist or clinician is unable to provide a definitive diagnosis and therefore warrant a needle or open biopsy.[53]

8.6.6 Postoperative MRI Findings

MRI can be invaluable in evaluating patients with new or recurrent symptoms following elbow surgery. Postoperative images can sometimes be difficult to interpret due to the challenge of differentiating between normal postoperative and pathologic changes. Orthopaedic implants can also create artifact that substantially distorts the image. However, both conventional MRI with a metal artifact reduction protocols and newer techniques such as Multi-acquisition Variable-Resonance Image Combination (MAVRIC) and Slice Encoding for Metal Artifact Correction (SEMAC) imaging can produce high-quality images even when metal is present.[8,55] After successful distal biceps repair, the distal biceps tendon normally appears heterogeneous and thickened.[56,57] MRI is useful to assess for complications such as heterotopic bone formation and graft rerupture. In patients with recurrent pain or difficulty throwing after UCL reconstruction, MR arthrography can be used to assess graft integrity and evaluate other potential issues such as posterior olecranon osteophytes, heterotopic ossification, and medial epicondyle avulsion.[58] MRI is also useful in patients with OCD of the capitellum after debridement and microfracture or with osteochondral mosaiplasty to assess the progress of healing and potentially assist with decision-making regarding return to full activity.[59,60] If the surgery is successful, the signal of the underlying bone should return to near-normal. After microfracture and debridement, the cartilage defect may or may not fill in with fibrocartilage, but some persistent thickening of the subchondral bone is common.[60] After mosaiplasty, as the graft heals, the signal intensity of the graft cartilage, which is initially hyperintense, will approach that of surrounding cartilage.[59] For patients with compressive neuropathy who have not improved after nerve release, MRI can evaluate for alternate causes of nerve compression, such as masses, and for structural or signal abnormalities within the nerve.

8.7 Common Clinical Questions

1. A 52 year-old male was water-skiing. The rope slipped. He struggled to keep his grip and felt a sudden painful pop in the elbow. He subsequently developed swelling and ecchymosis in the antecubital area. Two proton-density fat-suppressed MRI axial images are shown below. What is the diagnosis?
   a) Triceps rupture.
   b) MCL complex injury.
   c) Distal biceps rupture.
   d) Elbow hemarthrosis.

2. Which of the following is a normal variant?
   a) Pseudodefect of the radius.
   b) Pseudodefect of the olecranon.
   c) Pseudodefect of the capitellum.
   d) Pseudodefect of the trochlea.

3. A 61-year-old male has had 9 months of lateral elbow pain. He attributes his symptoms to his job as a cashier at a paint store, where he lifts dozens of heavy paint cans a day. He is now on light duty, but his symptoms have persisted. A T2 coronal MRI image is shown below. Which of the following examination findings is most likely?
   a) Pain with resisted wrist extension with the elbow extended.
   b) Pain with resisted wrist extension with the elbow flexed.
   c) Pain with supination.
   d) Varus instability.

4. A short tau inversion recovery (STIR) coronal MRI image is shown below. What is the most likely clinical scenario?
   a) A 45-year-old with overuse related medial elbow pain.
   b) A 22-year-old who dislocated his elbow playing football.
   c) A 64-year-old with rheumatoid arthritis.
   d) A 52-year-old with numbness and tingling in the small and ring finger.

5. Which of the following is most useful in ruling out septic olecranon bursitis?
   a) Well-defined margins.
   b) Bursal size.
   c) Bursal septation.
   d) Absence of rim and soft-tissue enhancement on postgadolinium T1-weighted images.

References

[1] Reichel LM, Milam GS, Sitton SE, Curry MC, Mehlhoff TL. Elbow lateral collateral ligament injuries. J Hand Surg Am. 2013; 38(1):184–201, quiz 201

[2] Zoner CS, Buck FM, Cardoso FN, et al. Detailed MRI-anatomic study of the lateral epicondyle of the elbow and its tendinous and ligamentous attachments in cadavers. AJR Am J Roentgenol. 2010; 195(3):629–636

[3] Anakwenze OA, Kancherla VK, Iyengar J, Ahmad CS, Levine WN. Posterolateral rotatory instability of the elbow. Am J Sports Med. 2014; 42 (2):485–491

[4] Mak S, Beltran LS, Bencardino J, et al. MRI of the annular ligament of the elbow: review of anatomic considerations and pathologic findings in patients with posterolateral elbow instability. AJR Am J Roentgenol. 2014; 203(6): 1272–1279

[5] Husarik DB, Saupe N, Pfirrmann CW, Jost B, Hodler J, Zanetti M. Ligaments and plicae of the elbow: normal MR imaging variability in 60 asymptomatic subjects. Radiology. 2010; 257(1):185–194

[6] Belentani C, Pastore D, Wangwinyuvirat M, et al. Triceps brachii tendon: anatomic-MR imaging study in cadavers with histologic correlation. Skeletal Radiol. 2009; 38(2):171–175

[7] Tagliafico A, Michaud J, Perez MM, Martinoli C. Ultrasound of distal brachialis tendon attachment: normal and abnormal findings. Br J Radiol. 2013; 86 (1025):20130004

[8] Johnson D, Stevens KJ, Riley G, Shapiro L, Yoshioka H, Gold GE. Approach to MR Imaging of the Elbow and Wrist: Technical Aspects and Innovation. Magn Reson Imaging Clin N Am. 2015; 23(3):355–366

[9] Wenzke DR. MR imaging of the elbow in the injured athlete. Radiol Clin North Am. 2013; 51(2):195–213

[10] Binaghi D. MR Imaging of the Elbow. Magn Reson Imaging Clin N Am. 2015; 23(3):427–440

[11] Potter HG. Imaging of posttraumatic and soft tissue dysfunction of the elbow. Clin Orthop Relat Res. 2000(370):9–18

[12] Jazrawi LM, Alaia MJ, Chang G, Fitzgerald EF, Recht MP. Advances in magnetic resonance imaging of articular cartilage. J Am Acad Orthop Surg. 2011; 19(7): 420–429

[13] Lomasney LM, Choi H, Jayanthi N. Magnetic resonance arthrography of the upper extremity. Radiol Clin North Am. 2013; 51(2):227–237

[14] Kaplan LJ, Potter HG. MR imaging of ligament injuries to the elbow. Radiol Clin North Am. 2006; 44(4):583–594, ixix.

[15] Skaggs DL, Mirzayan R. The posterior fat pad sign in association with occult fracture of the elbow in children. J Bone Joint Surg Am. 1999; 81(10):1429–1433

[16] O'Dwyer H, O'Sullivan P, Fitzgerald D, Lee MJ, McGrath F, Logan PM. The fat pad sign following elbow trauma in adults: its usefulness and reliability in suspecting occult fracture. J Comput Assist Tomogr. 2004; 28(4):562–565

[17] Kaas L, Turkenburg JL, van Riet RP, Vroemen JP, Eygendaal D. Magnetic resonance imaging findings in 46 elbows with a radial head fracture. Acta Orthop. 2010; 81(3):373–376

[18] Furushima K, Itoh Y, Iwabu S, Yamamoto Y, Koga R, Shimizu M. Classification of Olecranon Stress Fractures in Baseball Players. Am J Sports Med. 2014; 42 (6):1343–1351

[19] Pudas T, Hurme T, Mattila K, Svedström E. Magnetic resonance imaging in pediatric elbow fractures. Acta Radiol. 2005; 46(6):636–644

[20] Griffith JF, Roebuck DJ, Cheng JC, et al. Acute elbow trauma in children: spectrum of injury revealed by MR imaging not apparent on radiographs. AJR Am J Roentgenol. 2001; 176(1):53–60

[21] Melloni P, Valls R. The use of MRI scanning for investigating soft-tissue abnormalities in the elbow. Eur J Radiol. 2005; 54(2):303–313

[22] Wei AS, Khana S, Limpisvasti O, Crues J, Podesta L, Yocum LA. Clinical and magnetic resonance imaging findings associated with Little League elbow. J Pediatr Orthop. 2010; 30(7):715–719

[23] Dubberley JH, Faber KJ, Patterson SD, et al. The detection of loose bodies in the elbow: the value of MRI and CT arthrography. J Bone Joint Surg Br. 2005; 87(5):684–686

[24] Schwartz ML, al-Zahrani S, Morwessel RM, Andrews JR. Ulnar collateral ligament injury in the throwing athlete: evaluation with saline-enhanced MR arthrography. Radiology. 1995; 197(1):297–299

[25] Hill NB, Jr, Bucchieri JS, Shon F, Miller TT, Rosenwasser MP. Magnetic resonance imaging of injury to the medial collateral ligament of the elbow: a cadaver model. J Shoulder Elbow Surg. 2000; 9(5):418–422

[26] Cohen SB, Valko C, Zoga A, Dodson CC, Ciccotti MG. Posteromedial elbow impingement: magnetic resonance imaging findings in overhead throwing athletes and results of arthroscopic treatment. Arthroscopy. 2011; 27(10): 1364–1370

[27] O'Driscoll SW, Bell DF, Morrey BF. Posterolateral rotatory instability of the elbow. J Bone Joint Surg Am. 1991; 73(3):440–446

[28] Grafe MW, McAdams TR, Beaulieu CF, Ladd AL. Magnetic resonance imaging in diagnosis of chronic posterolateral rotatory instability of the elbow. Am J Orthop (Belle Mead NJ). 2003; 32(10):501–503, discussion 504

[29] Terada N, Yamada H, Toyama Y. The appearance of the lateral ulnar collateral ligament on magnetic resonance imaging. J Shoulder Elbow Surg. 2004; 13 (2):214–216

[30] Potter HG, Weiland AJ, Schatz JA, Paletta GA, Hotchkiss RN. Posterolateral rotatory instability of the elbow: usefulness of MR imaging in diagnosis. Radiology. 1997; 204(1):185–189

[31] LiMarzi GM, O'Dell MC, Scherer K, Pettis C, Wasyliw CW, Bancroft LW. Magnetic resonance arthrography of the wrist and elbow. Magn Reson Imaging Clin N Am. 2015; 23(3):441–455

[32] Hackl M, Wegmann K, Ries C, Leschinger T, Burkhart KJ, Müller LP. Reliability of magnetic resonance imaging signs of posterolateral rotatory instability of the elbow. J Hand Surg Am. 2015; 40(7):1428–1433

[33] Williams BD, Schweitzer ME, Weishaupt D, et al. Partial tears of the distal biceps tendon: MR appearance and associated clinical findings. Skeletal Radiol. 2001; 30(10):560–564

[34] Koplas MC, Schneider E, Sundaram M. Prevalence of triceps tendon tears on MRI of the elbow and clinical correlation. Skeletal Radiol. 2011; 40(5):587–594

[35] Kraushaar BS, Nirschl RP. Tendinosis of the elbow (tennis elbow). Clinical features and findings of histological, immunohistochemical, and electron microscopy studies. J Bone Joint Surg Am. 1999; 81(2):259–278

[36] Walton MJ, Mackie K, Fallon M, et al. The reliability and validity of magnetic resonance imaging in the assessment of chronic lateral epicondylitis. J Hand Surg Am. 2011; 36(3):475–479

[37] Potter HG, Hannafin JA, Morwessel RM, DiCarlo EF, O'Brien SJ, Altchek DW. Lateral epicondylitis: correlation of MR imaging, surgical, and histopathologic findings. Radiology. 1995; 196(1):43–46

[38] Qi L, Zhu ZF, Li F, Wang RF. MR imaging of patients with lateral epicondylitis of the elbow: is the common extensor tendon an isolated lesion? PLoS One. 2013; 8(11):e79498

[39] van Leeuwen WF, Janssen SJ, Ring D, Chen N. Incidental magnetic resonance imaging signal changes in the extensor carpi radialis brevis origin are more common with age. J Shoulder Elbow Surg. 2016; 25(7):1175–1181

[40] Boutin RD, Brossmann J, Sartoris DJ, Reilly D, Resnick D. Update on imaging of orthopedic infections. Orthop Clin North Am. 1998; 29(1):41–66

[41] Berquist TH. Musculoskeletal infection. In: Berquist TH, ed. MRI of the musculoskeletal system. 5th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2006

[42] Floemer F, Morrison WB, Bongartz G, Ledermann HP. MRI characteristics of olecranon bursitis. AJR Am J Roentgenol. 2004; 183(1):29–34

[43] Satake H, Takahara M, Harada M, Maruyama M. Preoperative imaging criteria for unstable osteochondritis dissecans of the capitellum. Clin Orthop Relat Res. 2013; 471(4):1137–1143

[44] Kijowski R, De Smet AA. MRI findings of osteochondritis dissecans of the capitellum with surgical correlation. AJR Am J Roentgenol. 2005; 185(6): 1453–1459
[45] Lubahn JD, Cermak MB. Uncommon nerve compression syndromes of the upper extremity. J Am Acad Orthop Surg. 1998; 6(6):378–386
[46] Keen NN, Chin CT, Engstrom JW, Saloner D, Steinbach LS. Diagnosing ulnar neuropathy at the elbow using magnetic resonance neurography. Skeletal Radiol. 2012; 41(4):401–407
[47] Bäumer P, Pham M, Ruetters M, et al. Peripheral neuropathy: detection with diffusion-tensor imaging. Radiology. 2014; 273(1):185–193
[48] Vucic S, Cordato DJ, Yiannikas C, Schwartz RS, Shnier RC. Utility of magnetic resonance imaging in diagnosing ulnar neuropathy at the elbow. Clin Neurophysiol. 2006; 117(3):590–595
[49] Kijowski R, Tuite M, Sanford M. Magnetic resonance imaging of the elbow. Part II: Abnormalities of the ligaments, tendons, and nerves. Skeletal Radiol. 2005; 34(1):1–18
[50] Chen FS, Diaz VA, Loebenberg M, Rosen JE. Shoulder and elbow injuries in the skeletally immature athlete. J Am Acad Orthop Surg. 2005; 13(3):172–185
[51] Khanna AJ, Cosgarea AJ, Mont MA, et al. Magnetic resonance imaging of the knee. Current techniques and spectrum of disease. J Bone Joint Surg Am. 2001; 83-A Suppl 2 Pt 2:128–141
[52] Cheng XG, You YH, Liu W, Zhao T, Qu H. MRI features of pigmented villonodular synovitis (PVNS). Clin Rheumatol. 2004; 23(1):31–34
[53] Frassica FJ, Khanna JA, McCarthy EF. The role of MR imaging in soft tissue tumor evaluation: perspective of the orthopedic oncologist and musculoskeletal pathologist. Magn Reson Imaging Clin N Am. 2000; 8(4):915–927
[54] Chen AL, Youm T, Ong BC, Rafii M, Rokito AS. Imaging of the elbow in the overhead throwing athlete. Am J Sports Med. 2003; 31(3):466–473
[55] Ariyanayagam T, Malcolm PN, Toms AP. Advances in metal artifact reduction techniques for periprosthetic soft tissue imaging. Semin Musculoskelet Radiol. 2015; 19(4):328–334
[56] Schmidt CC, Diaz VA, Weir DM, Latona CR, Miller MC. Repaired distal biceps magnetic resonance imaging anatomy compared with outcome. J Shoulder Elbow Surg. 2012; 21(12):1623–1631
[57] Alemann G, Dietsch E, Gallinet D, Obert L, Kastler B, Aubry S. Repair of distal biceps brachii tendon assessed with 3-T magnetic resonance imaging and correlation with functional outcome. Skeletal Radiol. 2015; 44(5):629–639
[58] Wear SA, Thornton DD, Schwartz ML, Weissmann RC, III, Cain EL, Andrews JR. MRI of the reconstructed ulnar collateral ligament. AJR Am J Roentgenol. 2011; 197(5):1198–1204
[59] Iwasaki N, Kato H, Kamishima T, Minami A. Sequential alterations in magnetic resonance imaging findings after autologous osteochondral mosaicplasty for young athletes with osteochondritis dissecans of the humeral capitellum. Am J Sports Med. 2009; 37(12):2349–2354
[60] Wulf CA, Stone RM, Giveans MR, Lervick GN. Magnetic resonance imaging after arthroscopic microfracture of capitellar osteochondritis dissecans. Am J Sports Med. 2012; 40(11):2549–2556

Answers to Common Clinical Questions

1. Answer: c. The first image shows the bicipital tuberosity, where the distal biceps tendon inserts onto the proximal radius. Normally, the distal biceps is visible here as low signal intensity tendon tissue. In this image, the bicipital tuberosity is bare as the distal biceps has ruptured and retracted proximally. The second image is a more proximal section showing the ruptured and retracted distal biceps tendon surrounded by high signal intensity fluid.
2. Answer: c. On sagittal MRI sequences of the elbow, a small depression of the posterior capitellum is sometimes seen. This can be a normal variant but is easily confused with an osteochondral fracture. Injuries of the posterior capitellum do occur, but they are uncommon. Marrow edema and other concomittant bony and/or ligamentus injuries can help distinguish between a traumatic defect and a pseudodefect.
3. Answer: a. The MRI shows interstitial tearing of the common extensor tendon, consistent with lateral epicondylitis. Lateral epicondylitis is caused by degenerative tearing of the extensor carpi radialis brevis tendon (ECRB). It is characterized by pain with wrist extension against resistance with the elbow extended, which stresses the ECRB.
4. Answer: b. The MRI shows a full-thickness tear of the ulnar collateral ligament with surrounding edema. Elbow dislocations usually result in multiple ligamentous and tendinous injuries though most often heal very well without surgery.
5. Answer: d. Nonseptic and septic olecranon bursitis can appear very similar on MRI. Poorly defined margins, marginal lobulation, bursal septation, heterogeneous internal signal, adjacent sot tissue edema, thickening/ edema of the triceps tendon, and localized bone marrow edema can be seen in both nonseptic bursitis. While rim and soft-tissue enhancement can also occur in both nonseptic and septic bursitis, when it is absent, septic bursitis can be ruled out.